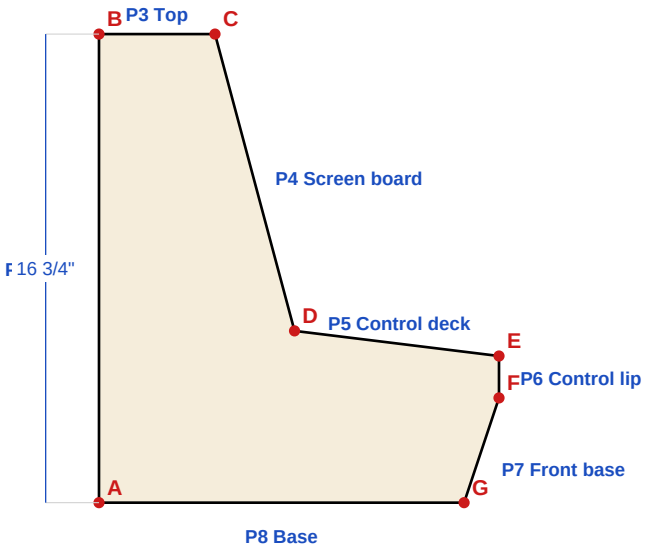
SUPERCADE - cardboard edition

Mini bartop. Printable plan set, built from the original Supercade CAD files.



P1 SIDE PANEL (x2) - side view, back on the left

AT A GLANCE

Overall height	16 3/4"
Overall depth	14 3/8"
Inside width	14 1/2"
Outside width	14 1/2" + 2 x side thickness
Control deck height	~5 3/4" above the table (7 deg tilt)
Screen	13.3" portable USB-C monitor
Material	cardboard body, 5 mm Foamex deck plate
Fasteners	kids' plastic cardboard screws + glue

WHAT'S IN THIS PACKET

1 This overview
2 Side panel - measure-and-mark drawing
3 Cut list (every part, every size)
4 Cross panels - fold-flange layout
5 Back panel - doors, cable hole + vents
6 Control panel layout + hole schedule
7 Shopping list
8 Assembly steps

1:1 print-and-tape templates in bartop/templates/:
control-panel-1to1.pdf
speaker-cutouts-1to1.pdf
side-panel-1to1.pdf

HOW THE CARDBOARD VERSION WORKS

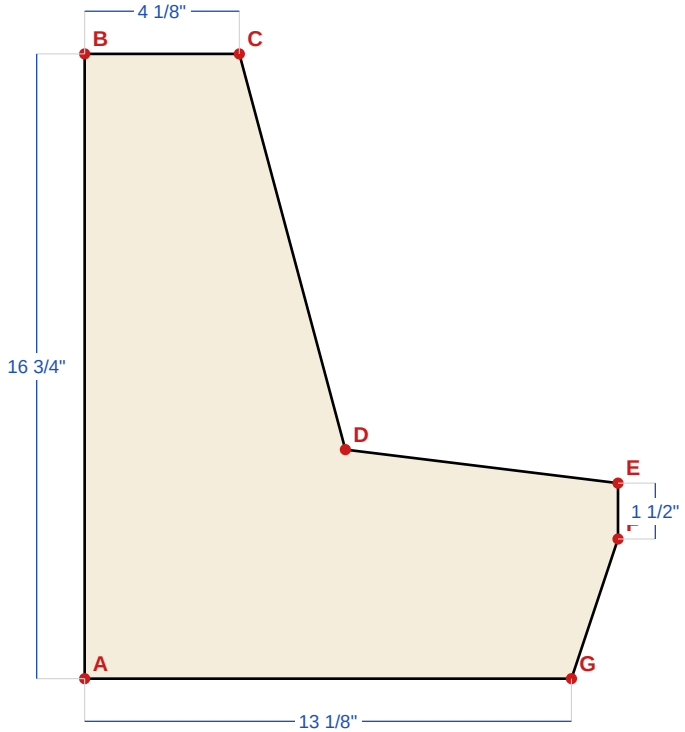
A short cabinet that stands on a table, OpenCade style: the screen and both speakers sit in ONE flat front board (P4) that leans back 15 degrees, so there's no recess to build. The deck is the control box's: same Foamex plate, same template, so it moves over with the wiring intact.

Every cross panel (P2-P8) gets a 1 1/2" flange folded down on each end. The flange lies flat against the inside of the side panel and a kit screw + nut goes through both layers, face to face.

The back has one big door for the monitor, speakers and computer, a cable hole and a row of vents (page 5).

P1 Side panel (make 2)

Draw it straight onto the cardboard with a tape measure and a long straightedge. Every edge is a straight line.



MARK THESE POINTS

Lay the sheet flat. Draw the BACK edge first as a long straight line (A to B). Every point is measured UP from the floor end along that line, then OUT square to it (use a set square or a big book).

Pt	Up from floor	Out from back	Is
A	0" (0.00)	0" (0.00)	table, back corner
B	16 3/4" (16.78)	0" (0.00)	top, back corner
C	16 3/4" (16.78)	4 1/8" (4.15)	top, front corner
D	6 1/8" (6.15)	7" (7.00)	screen board meets deck
E	5 1/4" (5.25)	14 3/8" (14.32)	deck, front edge
F	3 3/4" (3.75)	14 3/8" (14.32)	lip, bottom
G	0" (0.00)	13 1/8" (13.07)	table, front corner

Join the dots in order A-B-C-D-E-F-G-A with straight lines.

Check your work: D-E should be 7 3/8" long and C-D 11". If not, a point is off.

Second side: cut the first one out, flip it over, and trace it. The two sides are mirror images, so tracing the flipped panel gives you the correct other hand automatically.

Flutes UP AND DOWN (parallel to A-B): that direction carries the screen.

Stiffness: glue a second layer to each side (flutes crossed), OR glue 1 1/2" wide ribs of cardboard on the inside, keeping the 1 1/2" strip along every edge free for the flanges. See the screw-length note on page 7 before you pick.

Each side is 16 3/4" x 14 3/8". Trace the second from the first, flipped over.

Cut list

Cross panels are 14 1/2" between the sides + a 1 1/2" fold flange each end = 17 1/2" wide blanks.

Part	Name	Qty	Cut size (L x W)	Edge	Notes
P1	Side panel	2	16 3/4" x 14 3/8"	-	Shape on page 2. Mirror pair.
P2	Back	1	16 3/4" x 17 1/2"	A-B	Cut a 10 1/2" x 11 1/2" rear door, a 1 1/2" cable hole and 4 vent holes (page 5).
P3	Top	1	4 1/8" x 17 1/2"	B-C	
P4	Screen board	1	11" x 17 1/2"	C-D	The whole front, one flat piece. Centred window 11 5/8" x 6 1/2", 1" to 7 1/2" down from the C edge. Two 2" round speaker holes 9 1/2" down (speaker template).
P5	Control deck	1	7 3/8" x 17 1/2"	D-E	1 layer of cardboard + the Foamex plate (P12) on top. Under each joystick, cut an opening just bigger than the stick's plate, so the stick bolts to the Foamex only.
P6	Control lip	1	1 1/2" x 17 1/2"	E-F	Front face under the players' wrists.
P7	Front base	1	4" x 17 1/2"	F-G	Leans in under the deck, so knees and fingers clear it.
P8	Base	1	13 1/8" x 17 1/2"	G-A	Stands on the table. The Pi and the ballast go on it.
P10	Monitor ledge	1	2 1/2" x 5 1/2"	-	Fold lengthwise into a 1 1/4" + 1 1/4" L. Glue one leg to the back of P4 right under the window, between the speakers; the monitor stands on the other. Two layers, flutes crossed.
P12	Deck top plate	1	7 3/8" x 14 1/2"	-	5 mm Foamex (PVC foam board). Tape the 1:1 template to it, cut every hole, then drill through into P5.
P13	Corner strips	8	1 1/2" x 14 1/2"	-	Folded into L-angles and glued into inside corners that feel loose.

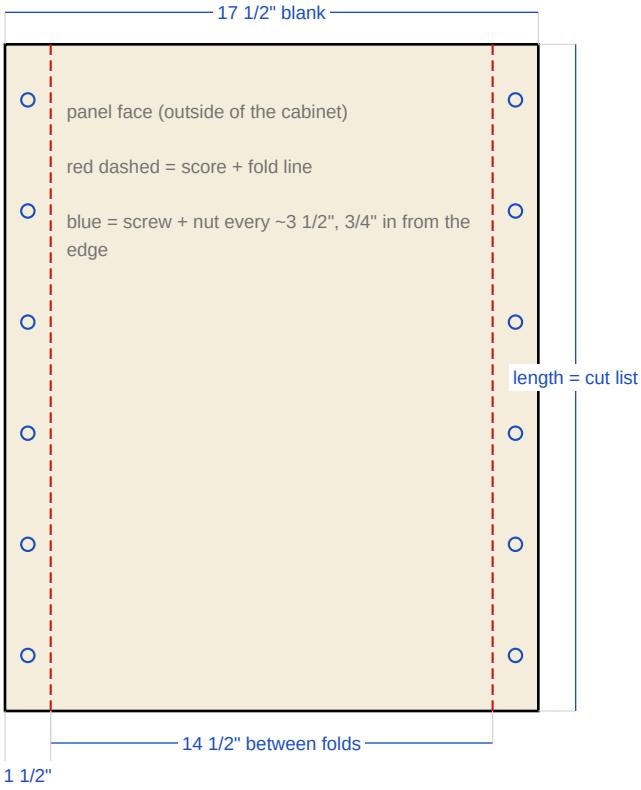
Cardboard: 2-3 big boxes. The longest parts (sides, back and screen board) are under 18" long.

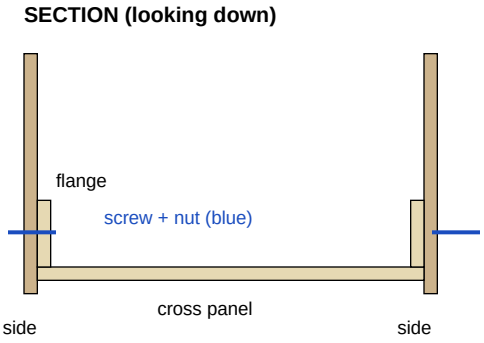
Every size is measured on the CAD edge. Cut cross panels 1/4" long and trim during the dry fit: cardboard is thicker than the pencil line, so corners will need a nibble.

Flutes: on the cross panels, run them ACROSS the 17 1/2" width so the flange fold goes along the flutes (cleaner fold, stiffer span).

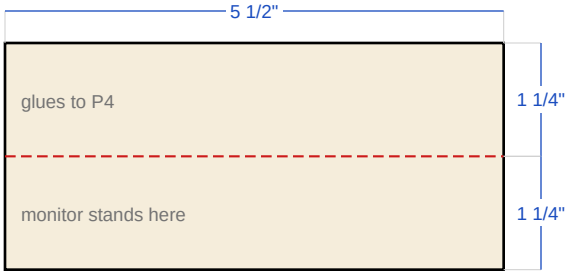
Cross panels - fold-flange layout

The same pattern for P2-P8. Score, fold 90 degrees, screw to the side panels.





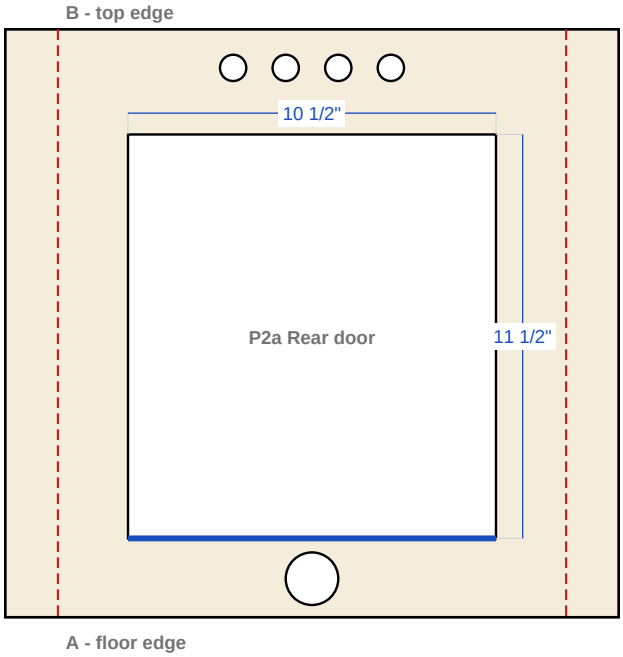
P10 MONITOR LEDGE



Cut two 5 1/2" x 2 1/2" strips and glue them together with the flutes crossed. Score the red line and fold to 90 degrees. Glue the top leg flat to the back of P4, its fold right at the bottom edge of the window, centred between the speaker holes. The monitor stands on the other leg, face against the window; a Velcro strip across its top holds it to P4. It only has to carry a 0.7 kg portable monitor, so no shelf is needed.

P2 Back panel - doors, cable hole + vents

Cut these before the back goes on. Measured on the outside face: across from the left fold, up from the floor (A) edge.



WHERE EACH ONE GOES

P2a Rear door: 10 1/2" wide x 11 1/2" tall, centred, bottom edge 2 1/4" up. Hinged along the bottom, Velcro at the top.

Cable hole: 1 1/2" circle, centred, 1 1/8" up to its middle.

Vents: 4 holes of 3/4", 1 1/2" apart, centred, 15 5/8" up to their middles.

HOW TO CUT THEM

Doors: cut three sides and score the hinge side (thick blue line) so the piece you cut out folds down and becomes the door. Glue a 1 1/2" strip of cardboard round the inside of the opening, like a picture frame: the back panel is what keeps the cabinet square, and the frame gives that back. Velcro at the top keeps it shut.

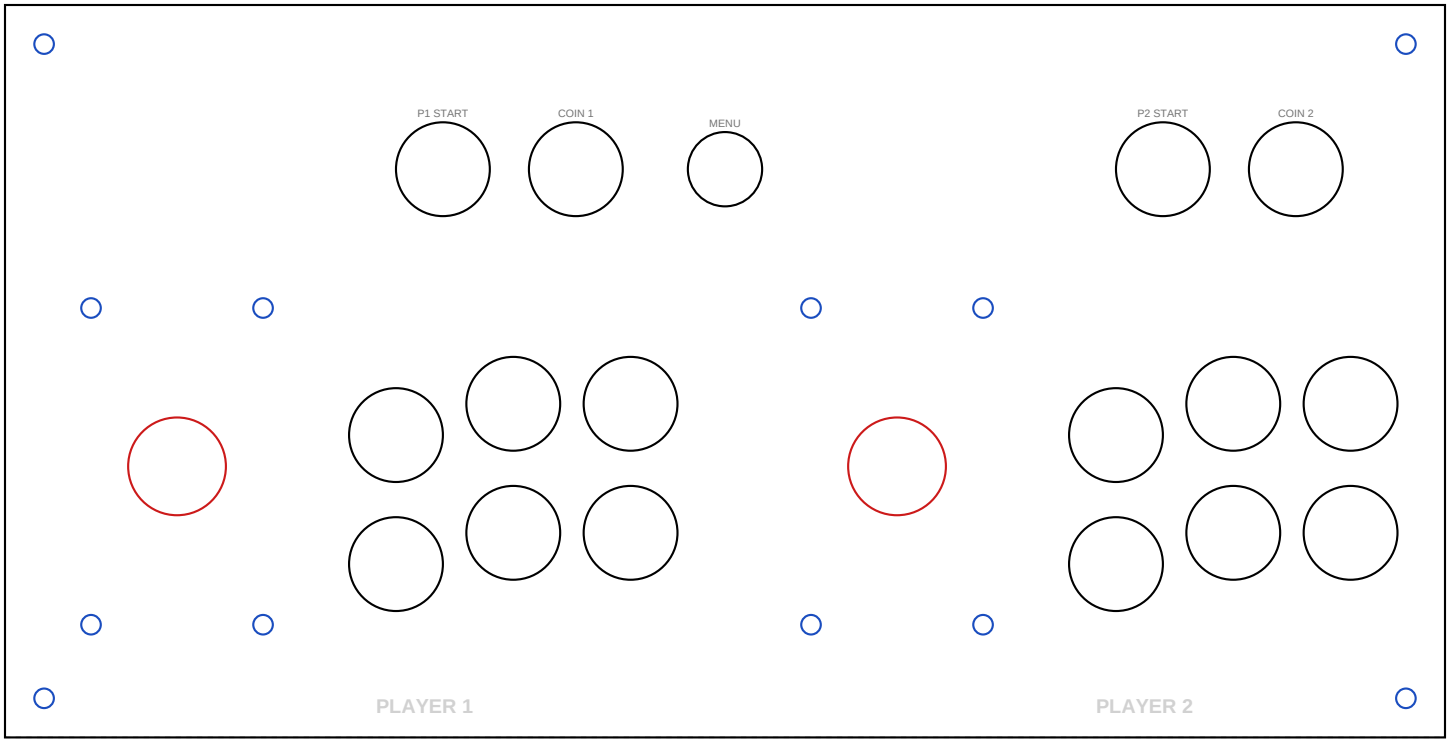
Cable hole: one power lead (or the controllers' USB leads) comes out here. Wrap tape round the edge so it doesn't fray. Plug everything inside into one power strip.

Vents: warm air rises, so they sit at the top. Don't cover them.

Leave at least 2" of solid board between any opening and the 1 1/2" fold flanges.

P5 Control panel layout

compact 2-player layout for the scaled deck. Print templates/control-panel-1to1.pdf for the real-size version.



dashed = full deck P5 (14 1/2" x 7 3/8"). Template is the whole deck: line its edges up with P5. Players stand at the bottom edge.

HOLE SCHEDULE

2	**25 mm**	Joystick shafts. The stick's plate bolts on underneath the deck.
12	**24 mm**	Action buttons. Use SCREW-IN buttons with a plastic nut - snap-in buttons only grip thin metal/acrylic.
2	**24 mm**	Coin buttons (Select on most emulators).
2	**24 mm**	Player Start buttons.
1	**19 mm**	Aux: menu / exit / volume.
12	**5 mm**	Bolt holes: joystick plates (4 per stick - CHECK against your own stick, spacing varies) and deck corners.

CUTTING ROUND HOLES

Tape the 1:1 template down. Push an awl (or the screw kit's punch tool) through every crosshair. Remove the paper, draw each circle with a compass, then cut with a sharp craft knife in several light passes. Faster and neater: a hole saw or step drill bit of the right size, run by a grown-up.

Cut small first and open up: the button should need a firm push, and the plastic nut does the rest. Keep the deck thin where the controls go (about 10 mm max): screw-in buttons and joysticks are made for thin panels.

Shopping list

Search terms, not specific listings - check sizes and reviews before you buy.

CARDBOARD + FASTENING

☐ **Big cardboard boxes**	2-3 large ones. The longest parts are under 18" long, so most moving boxes work.
☐ **Kids' cardboard construction kit with screws + nuts**	e.g. a 400-piece kit with safe saw + crease roller. ~80 screws get used, so one kit is plenty.
☐ **5 mm PVC foam board ("Foamex"), A3 / 12 x 18" is enough**	For the control deck top plate. Cuts with a craft knife, holds buttons firmly.
☐ **PVA / white glue + hot glue gun**	Glue the laminations and the box-beam shelves.
☐ **Nylon or steel M4/M5 x 20 mm bolts + nuts + washers**	8 for the joysticks, 4 for the deck plate corners.
☐ **Wide packing tape / gaffer tape**	Edge-protecting the corners and the bottom edge.
☐ **Velcro strips**	Holds the back door shut and the top of the monitor to the screen board.

CONTROLS + ELECTRONICS

☐ **"Zero delay" 2-player arcade kit with 24 mm buttons**	Needs 2 sticks + 16 x 24 mm buttons (12 action + 2 start + 2 coin). If the kit only has 30 mm, buy a 24 mm pack. SCREW-IN type.
☐ **13.3" portable USB-C monitor**	~0.7 kg. Built-in speakers are common, so check and skip separate ones.
☐ **Computer**	A Raspberry Pi 4/5 on the base, or run the controls out of the cable hole to a laptop or TV.
☐ **Small speakers (optional)**	Only if the monitor has none. Two 2" (50 mm) round drivers + a small USB amp; the cut-outs are in the templates.
☐ **Weight + grip**	About 1 kg (a bag of rice) on the base, and a non-slip mat under it.

TOOLS

☐ **Metal ruler (1 m / 40") + set square**	For drawing the side panels.
☐ **Craft knife + cutting mat**	Adults do the long cuts and the Foamex. Kids use the kit's safe saw.
☐ **Crease roller (in the kit) or a blunt butter knife**	For scoring fold lines.
☐ **Compass, pencil, tape measure**	Or a drill with 24 mm + 25 mm hole saws / a step bit for neat holes.

SCREW LENGTH - READ BEFORE YOU LAMINATE

Kids' plastic cardboard screws are short: they're made to join two layers of ordinary box cardboard. Before cutting anything, screw a scrap of your flange cardboard to a scrap of your side-panel cardboard, nut on. If it won't reach through a laminated (2-layer) side, keep the sides single-layer and stiffen them with glued ribs instead (page 2), or use nylon nuts and bolts at those joints.

Assembly steps

Dry-fit with screws only, glue once everything lines up.

1. **Test the screws**
   Screw scrap to scrap in the exact layers you'll use (page 7 note).

2. **Draw + cut the sides (P1)**
   Page 2. Cut one, flip it, trace the second. Label the insides 'LEFT' and 'RIGHT'.

3. **Cut + score the cross panels (P2-P8)**
   Page 3 sizes, page 4 flanges. Cut the back door, cable hole and vents in P2 now (page 5), and the window + speaker holes in P4.

4. **Build the control deck (P5 + P12)**
   Cut every hole in the Foamex plate from the 1:1 template, lay it on P5 and mark through. Cut P5's holes plus a bigger opening under each stick. Bolt plate + P5 together at the corners, then mount sticks and buttons on the bench.

5. **Box it up**
   Lay one side flat, inside face up. Screw in P8 (base), P2 (back) and P3 (top), then P7, P6 and the pre-wired deck P5. Put the second side on top and screw everything.

6. **Screen board + ledge (P4, P10)**
   Glue the P10 ledge to the back of P4 under the window, between the speaker holes. Screw P4 in and fit the speakers behind their holes.

7. **Monitor + electronics**
   Through the back door: stand the monitor on the ledge, face against the window, and Velcro its top to P4. Encoders and the Pi go on the base; one lead out of the cable hole.

8. **Glue + finish**
   Once it all works, back out one screw at a time, run glue into the joint, re-screw. Tape the edges, then paint or wrap it.

SAFETY. Light, but it tips forward if someone leans on the deck: put 1 kg on the base, a non-slip mat under it, and keep it back from the table edge. Adults do the knife and Foamex cutting; kids use the kit's safe saw. Keep the vents clear.